



You2.0: A Multi-Modal Machine Learning Framework for Personalized Lifestyle Monitoring and Preventive Health Intelligence

RG 71 - Aashi Gupta[221030140], Lakshay Malik[221030022]

Department of Computer Science and Engineering

Under the supervision of Dr. Monika [Assistant Professor Senior Grade]

Problem Statement

- Sedentary lifestyle and poor habits increase health risks.
- Existing apps track limited wellness metrics.
- Most systems lack personalized health insights.
- Need for unified lifestyle monitoring and prevention system.

Objectives

- To create a system that records diet, sleep, steps, hydration, and mood, either manually or through sensors.
- To apply machine learning models, which will recognise the user's mood and the food they ate.
- To provide personalized and real-time recommendations to promote healthy habits.
- To provide reference reports weekly or monthly to the consumer with trends analysis.

Methodology

- Collected sensor and user-input lifestyle data
- Trained food and mood recognition models
- Processed wellness metrics through AI engine
- Generated recommendations and health insights
- Stored data for trend analysis and monitoring

Implementation

- Developed You2.0 mobile application with React Native and Expo for cross-platform wellness monitoring.
- Used pedometer and manual entry to track steps, sleep, hydration, sedentary hours, and activities completed daily.
- Used EfficientNet-B0 on Food-101 dataset and FER-2013 dataset for food recognition, calorie estimation and mood detection.

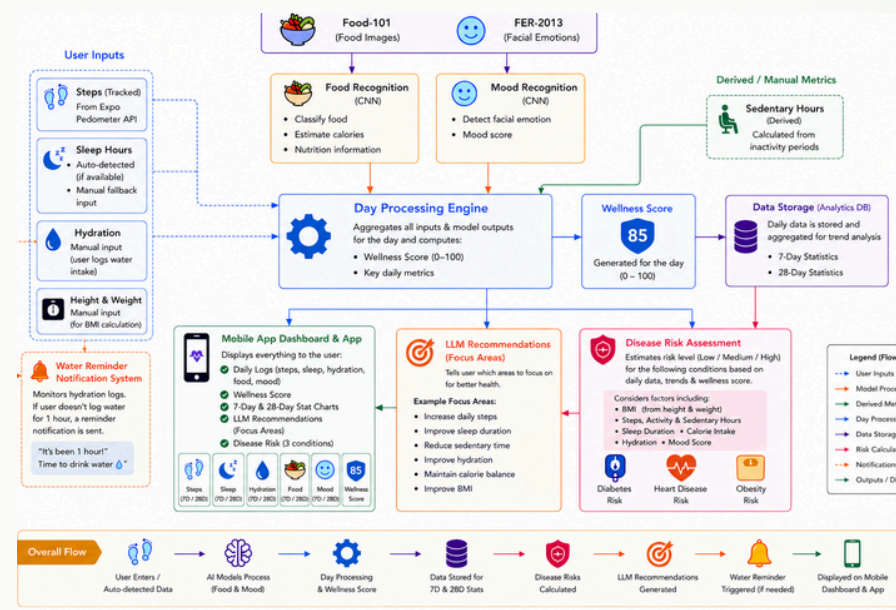


Fig 1: Design and System Architecture

- Built Multi-modal wellness engine that combines lifestyle metrics to produce a wellness score and evaluate the risk for disease.
- Developed LLM-based recommendation to provide insights into health-related issues such as Diabetes, Obesity and Heart Disease Risk.
- Created Dashboard analytics, provide weekly/monthly trend visuals, and give smart hydration reminder notifications.

Results

Model	Accuracy
Food recognition	86% accuracy
Mood recognition	70% accuracy

- Real-time wellness monitoring achieved
- AI-based personalized recommendations generated
- Interactive dashboard successfully implemented
- Lifestyle trends visualized successfully

Conclusion

You2.0 successfully integrates food recognition, mood detection, activity monitoring, hydration tracking, smart notifications and AI-based recommendations into a unified wellness platform for preventive healthcare.

Future Work

- Introduce gamified habit tracking.
- Add personalized diet plans.

References

- [1] Bossard, L., Guillaumin, M., & Van Gool, L. (2014, September). Food-101—mining discriminative components with random forests. In European conference on computer vision (pp. 446-461). Cham: Springer International Publishing.
- [2] Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., ... & Bengio, Y. (2013, November). Challenges in representation learning: A report on three machine learning contests. In International conference on neural information processing (pp. 117-124). Berlin, Heidelberg: Springer berlin heidelberg.
- [3]Singha, P. S., Panda, B., Firdaus, S. B., & Ghosh, D. (2025). Machine Learning (ML) Techniques in Healthcare Systems: A Mini Review. Recent Patents on Engineering, 19(7), E220224227338.